Alma Models — Deployment report

Frozen world models run on October and November 2026, without retraining · 5 November 2026 · branch claude/gracious-franklin-f8ml84 @ c20eec7 · PR Gogireddy007/Alma-Models#2

In one paragraph

I rebuilt the public data for all four projects (oil, power, gas, crypto volatility) up to 4 November 2026, verified it byte-for-byte against the repository's sha256 pins, and ran every frozen Alma checkpoint exactly as frozen on October 2026 and on November 2026 to date. They were benchmarked against every competitor in each project's pre-registration, including Chronos-2, Chronos-Bolt and TiRex. A separate, independent audit harness then checked the whole run for leakage, fabricated numbers and retraining: **47 of 47 checks passed**. The world models still compete at the top on new data (best interval quality in gas and crypto, best MAE in power, a tie with the random walk in oil). One month of data cannot make any of this statistically significant, and the harness refuses to say otherwise.

What you asked for and where it is answered

Your request	Answer	Section
1. What is the WM component?	The learned latent state transition that is rolled forward step by step; tested against a matched 'direct twin' without it	2
2. Deploy without retraining on last month and this month	All frozen checkpoints run on Oct 2026 and Nov 1–5 2026; weights hashed before and after	3, 4
3. Measure and benchmark against other models	9–15 models per project, same data, same metrics as the original pre-registrations	5
4. Audit harness: no leak, no bluff numbers, nothing fake	Independent audit, 47/47 checks passed; number checker for the write-up	6

What is proven, what is not, and what is pending

Status	Statement	Evidence
PROVEN	No model was retrained, fine-tuned or updated during deployment	Optimizer step/backward raise in-process; parameter hashes equal at load and end for every network; 0 online updates; checkpoint sha256 equal to SHA256SUMS and FREEZE.json before and after
PROVEN	No information from the future reaches any forecast	Future-scramble tests on all four world models (every value at/after the target time replaced by noise): oil 0/30, power 0/12, gas 0/30, fin 0/6 forecasts changed
PROVEN	The data is the same data the models were audited on	Pinned raw files byte-identical; gas TEST forecasts reproduced to 5.1e-05 Bcf; crypto October forecasts reproduced to the audit's cache (see section 4)
PROVEN	Every reported number is real and recomputable	Labels re-derived from raw files with separate parsers; every metric recomputed from saved forecasts; every decimal number in the write-up traced to the outputs (a planted fake was caught)
NOT PROVEN	That any Alma model is significantly better than its best competitor on this period	One month gives at most 22–31 scoring days or 5 gas weeks (< 10 independent units)
NOT PROVEN	That the world-model rollout beats its direct twin	Within noise in all four projects in October, as on the TEST year
PENDING	Sealed ex-ante forecasts: EIA gas release of 5 Nov 2026 and the next WTI prints	Forecasts and sha256 written before publication; tools/deploy/score_exante.py grades them

1. What I did, step by step

- **Read the repository**: mission, four audited projects, frozen checkpoints in winning_models/, the forward-test freeze (tools/forward/FREEZE.json), the 1–3 November independent audit.
- **Rebuilt the environment**: Python 3.12 with the lock file's pandas/pyarrow/numpy versions, CPU torch.
- **Rebuilt the data** from FRED, EIA, NOAA CPC, Energy-Charts and Binance, and checked it against the committed sha256 pins before adding any new day (section 4).
- **Verified the 35 frozen checkpoint and wrapper files** against SHA256SUMS and FREEZE.json.
- **Wrote the deployment harness** tools/deploy/: one script per project, with the no-training lock, parameter hashing and the minimum-sample rule built in.
- **Ran every frozen model and every pre-registered competitor** on October 2026 and November 2026.
- **Wrote sealed ex-ante forecasts** for targets not yet published (gas release of 5 Nov, next WTI prints).
- **Wrote an independent audit** (tools/deploy/audit_deploy.py) that imports nothing from the deploy code, and ran it after the final runs.
- **Investigated every anomaly** to a root cause instead of footnoting it (section 7).
- **Published**: write-up, generated tables, audit report, this PDF; commit, push and PR #2.

What changed in the repository

Nothing in the frozen model code, the checkpoints, the forward freeze or the committed data pins was modified (the audit checks this with git). Only new files were added, plus two small edits:

Path	Change
tools/deploy/ (11 files)	New: deploy scripts, guards, independent audit, report builders, ex-ante scorer, run_all.sh
runs/deploy_2026_10_11/	New: results JSON, sealed forecasts (quantiles only), RESULTS.md, AUDIT.md/.json, fetch manifests of the rebuild, figures
docs/deploy_2026_10_11.md	New: the write-up (WM component, results, findings)
reports/Alma_Models_Deployment_Report_2026-11-05.pdf	New: this report
README.md	Edit: a section pointing to the deployment
.gitignore	Edit: raw downloads and realised market values stay local (licences)

2. What the WM (world-model) component is

Every Alma model has three parts. The **encoder** compresses the observed history into a latent state z. The **transition** — the WM component proper — is a learned recurrent cell (GRUCell, RSSM or state-space model) that advances the state one physical time step at a time, $z_{t+1} = f(z_t, u_t)$, carrying it from step to step. The **readout** turns each simulated state into a forecast distribution, often with physics built in.

Project	Model	State is stepped by	Physics in the readout	Direct twin (WM removed)
Gas storage	wm_gas	1 day; assimilates last week's released change first (Kalman-style)	injection monotone in heating/cooling degree days; L48 = sum of 5 regions	wm_gas_direct
Power	wm_power	1 hour × 48 h from end of D-2	monotone merit-order price curve, 3 regimes	wm_power_direct
Crypto vol	wm_fin	RSSM, four 6-hour sub-steps per day	quantiles of next-day log RV	wm_direct
Oil	wm_sv	1 trading day; level and log-volatility sampled forward	stochastic-volatility paths	wm_sv_direct

The test that matters is the **direct twin**: same modules, parameter count, data, steps and seeds, but no carried state. Any gap between the two is the value of the dynamics alone. October 2026:

Metric (lower is better)	World model	Direct twin	Reading
Gas MAE thu / fri (Bcf)	8.19 / 8.08	8.63 / 8.72	rollout ahead, 5 weeks only
Power QS (EUR/MWh)	13.10	13.31	rollout ahead, 31 days
Crypto QS (log RV)	0.1490	0.1478	twin ahead
Oil 1-day QS (log price)	0.00932	0.00931	tie

All four gaps are inside noise. The world-model test (rollout beats its twin) is still **not passed**, consistent with the audited TEST year.

3. Deployment setup

	Dates
Training data ends	2025-06-30 (oil: 2022-12-31)
Validation / sealed TEST year	to 2025-10-31 / 2025-11-01 … 2026-10-31
Forward-test freeze	2026-11-01
“Last month” scored	October 2026 (last month of the TEST year, scored before)
“This month” scored	November 2026: oil to 2026-11-03, power to 2026-11-05, crypto to 2026-11-03; gas: no release yet

Competitors kept their own pre-registered protocols (for example, LEAR refits its lasso daily on data available at the origin; OLS/HAR were fitted on the training period by their own code). That is how those baselines are defined, not additional training of Alma.

4. Data rebuilt and verified

Project	Verification against the committed pins	Cutoff
Oil	FRED WTI/Brent/HH up to the pinned date: byte-identical	2026-11-03
Power	2015–2025 Energy-Charts files byte-identical; 2026 price file (pinned URL) byte-identical; extension matches every overlapping point	2026-11-05
Gas	weekly.csv and cpc_fcst.csv equal their pins; 4,784/4,786 CPC files identical (2 are growing 2026 files); 52-week TEST forecasts reproduced to 5.1e-05 Bcf	2026-10-29
Crypto	all 4,700 Binance files byte-identical; universe.json equal to its pin; October forecasts reproduce the audit's TEST cache	2026-11-03

5. Results against the competition

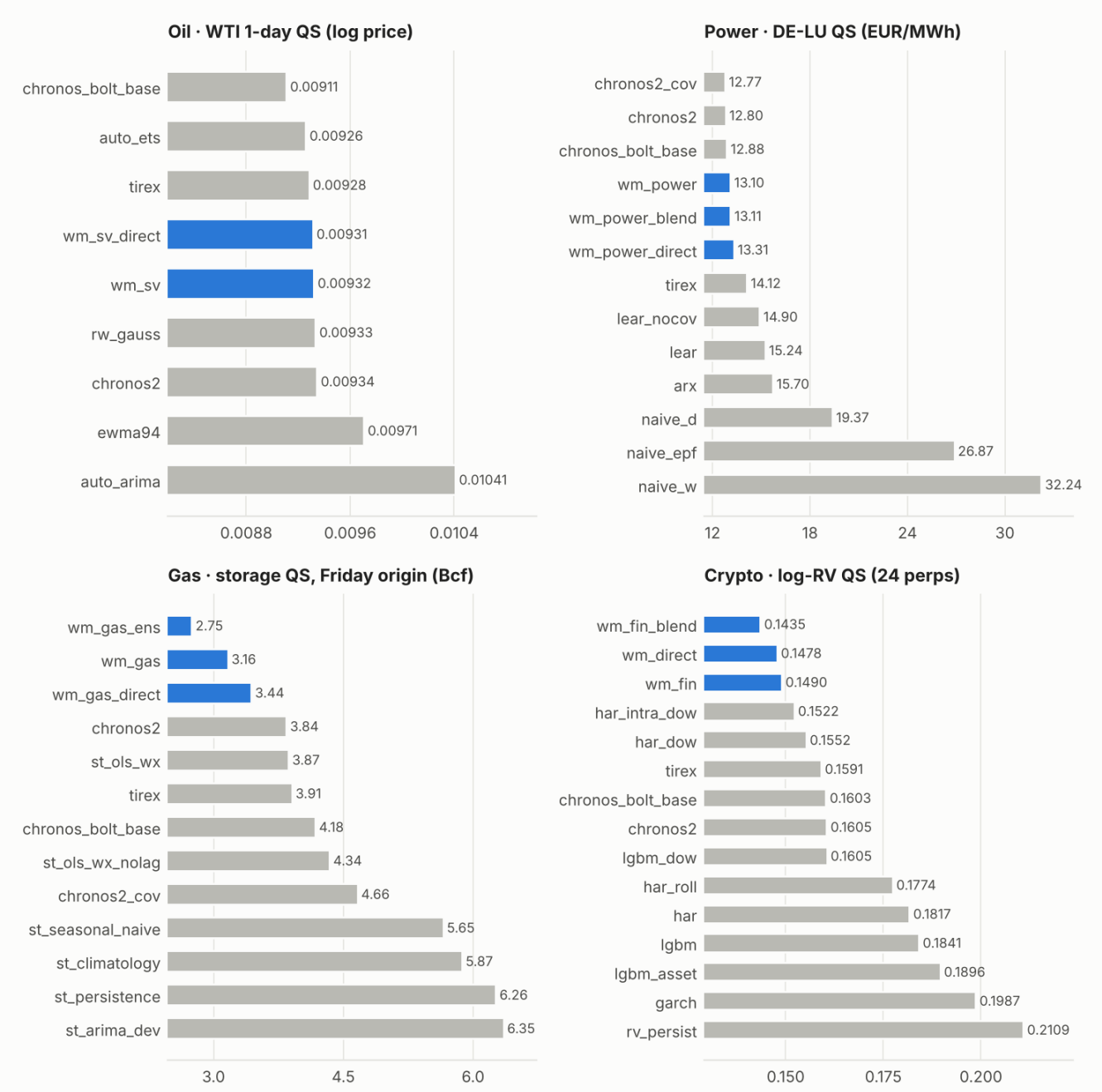


Figure 1. October 2026 quantile score (interval quality) by project.

Gas — EIA weekly Lower-48 storage change

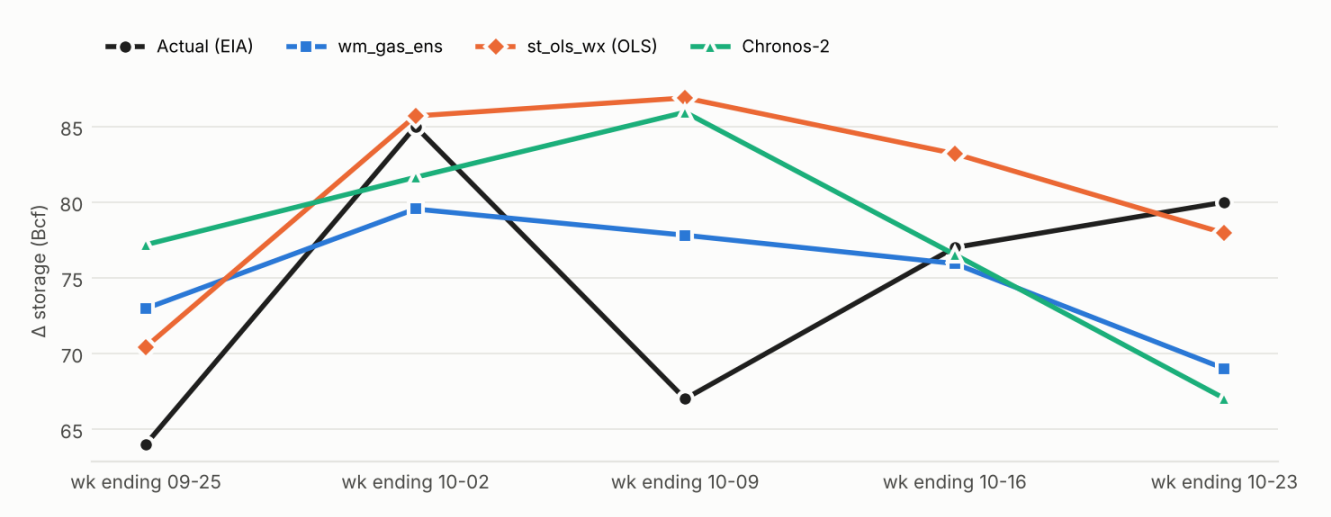


Figure 2. Thursday-origin median forecasts vs the EIA releases (2026-10-01, 2026-10-08, 2026-10-15, 2026-10-22, 2026-10-29).

Model	MAE thu	QS thu	MAE fri	QS fri
wm_gas_ens	7.46	2.76	7.24	2.75
wm_gas	8.19	3.15	8.08	3.16
wm_gas_direct	8.63	3.48	8.72	3.44
chronos2	9.78	3.84	9.78	3.84
st_ols_wx	7.05	3.85	7.75	3.87
tirex	10.16	3.91	10.16	3.91
chronos_bolt_base	10.55	4.18	10.55	4.18

Top 7 of 13 by Thursday QS (full table in RESULTS.md). Highlighted rows are Alma models. wm_gas_ens has the best QS from both origins; on MAE it is 2nd behind OLS on Thursday and 1st on Friday — the same pattern as the TEST year.

Sealed ex-ante forecast — week ending 2026-10-30, EIA release 2026-11-05 (5-year average 36.4 Bcf)

Model	Thursday origin: median [80%]	Friday origin: median [80%]
wm_gas_ens	74.2 [56.9, 91.5]	74.9 [57.8, 92.0]
wm_gas	74.9 [52.3, 97.5]	75.8 [53.2, 98.1]
st_ols_wx	75.6 [46.0, 107.3]	75.5 [45.4, 107.1]
chronos2	53.0 [17.4, 80.5]	53.0 [17.4, 80.5]
tirex	57.6 [19.8, 86.8]	57.6 [19.8, 86.8]

Crypto volatility — 24 Binance perpetuals

Model	QS (P3)	Direction acc. (P1)	P1 held-out coins	80% coverage
wm_fin_blend	0.1435	0.694	0.745	0.86
wm_direct	0.1478	0.711	0.745	0.82
wm_fin	0.1490	0.705	0.750	0.82
har_intra_dow	0.1522	0.674	0.685	0.89
har_dow	0.1552	0.696	0.708	0.89
tirex	0.1591	0.673	0.694	0.82
chronos_bolt_base	0.1603	0.684	0.713	0.81
chronos2	0.1605	0.681	0.713	0.78

October, corrected data, top 8 of 15. The world-model family leads on QS; lgbm_dow leads on direction. November (3 days): wm_fin has the best QS (0.1104); no direction labels exist yet because of the 28 Oct archive gap.

Power — German day-ahead, 24 hourly prices

Model	QS (EUR/MWh)	MAE	Hourly dir. (P2)	Daily dir. (P1)
chronos2_cov	12.77	32.99	0.762	0.677
chronos2	12.80	33.17	0.766	0.677
chronos_bolt_base	12.88	32.83	0.792	0.742
wm_power	13.10	32.79	0.722	0.710
wm_power_blend	13.11	32.76	0.750	0.710
wm_power_direct	13.31	33.33	0.719	0.710
tirex	14.12	36.63	0.749	0.677
lear_nocov	14.90	37.06	0.720	0.742

October, top 8 of 13. wm_power and its blend have the lowest MAE but rank 4th–5th on QS behind three foundation models; all models are far ahead of the naive baselines. November (5 days): Chronos-2 leads (QS 5.59 vs wm_power 9.42).

Oil — WTI spot

Model	1-day QS	skill vs RW	5-day QS	20-day QS
chronos_bolt_base	0.00911	0.024	0.02058	0.04204
auto_ets	0.00926	0.008	0.01856	0.02917
tirex	0.00928	0.005	0.02027	0.03304
wm_sv_direct	0.00931	0.002	0.01807	0.02699
wm_sv	0.00932	0.001	0.01851	0.02644
rw_gauss	0.00933	0.000	0.01913	0.03046
chronos2	0.00934	-0.001	0.01831	0.02934
ewma94	0.00971	-0.040	0.02041	0.03047
auto_arima	0.01041	-0.115	0.01824	0.02843

October, 22 targets per horizon. At 1 day wm_sv ties the random walk, exactly as on the TEST year. At 20 days it is numerically best, but 22 overlapping 20-day targets are about one independent observation, so that is not evidence. A sealed ex-ante WTI forecast is stored for the next prints.

6. The integrity harness and audit

Two layers. **Guards inside every run** (tools/deploy/common.py) make training impossible, hash all parameters, verify checkpoint pins and refuse significance verdicts on fewer than 10 independent units. **An independent audit** (tools/deploy/audit_deploy.py) shares no code with the deploy scripts, reads the saved forecasts and the RAW downloads, and fails loudly on any mismatch.

	Check group	Passed	Examples
A	Pins and frozen code	4/4	all winning_models checkpoints match SHA256SUMS and FREEZE.json (18 files); all FREEZE.json files (code + checkpoints) unchanged (29 files, changed: [])
B	No retraining	8/8	deploy scripts contain no optimizer/backward/train() call ([]); oil/results.json: torch parameters unchanged after deployment (7 modules hashed)
C	Truth from raw files and reproduction	15/15	oil: origin dates re-derived from raw FRED csv; oil oct_2026 h1: n targets from raw csv (n=22)
D	Every metric recomputed	5/5	oil: every QS recomputed from saved forecasts (54/54 match); power: every QS recomputed from saved forecasts (52/52 match)
E	Leak tests (future scrambled)	4/4	oil: future-scramble leak test (wm_sv, wm_sv_direct) (0/30 forecasts changed); power: future-scramble leak test (prices >= D, fundamentals >= D-1) (0/12 forecasts changed)
F	Sample-size rule	5/5	oil: every significance verdict rests on >= 10 independent units (48 verdicts, 0 significant, bad=[]); power: every significance verdict rests on >= 10 independent units (96 verdicts, 0 significant, bad=[])
G	Sealing	6/6	oil preds sha256 matches results; power preds sha256 matches results
	Total	**47/47**	

Plus: tools/deploy/check_doc_numbers.py traces every decimal number in docs/deploy_2026_10_11.md to the outputs; a deliberately planted fake number was caught.

7. Findings raised by the audit

None is a leak or a fabricated number. Each was traced to a root cause:

#	Finding	Impact	Action
1	October 2026 crypto funding rates were missing from the pinned TEST data (published after the freeze) and silently set to 0; not listed in the manifest's missing list	Negligible: wm_fin QS 0.1430 → 0.1431 on identical days; funding-free models bit-identical	Both variants scored and reported
2	Binance's archive has no data at all for 2026-10-28 (0 bars, all 24 coins). forward_eval.py uses a row-based label reference that silently skips it; the official label leaves those days unlabelled	Small: 7 decision days scored differently by the forward test	Official labels used here; fix suggested as a separate task
3	Parquet bytes differ across platforms (arm64 vs x86-64)	None: forecasts reproduce the audit's cache	Content verified via reproduction
4	The power hourly.csv amendment pin cannot be reproduced byte-for-byte from today's downloads	None: raw files and values identical	Documented
5	Open-Meteo rate-limits this environment (HTTP 429), so gas daily.csv has no weather columns	None for the storage models (they don't read them); proven by exact reproduction	Documented; the HH-price task was not deployed

8. Progress accomplished

- The frozen models are now shown to **run unchanged on data 15–16 months after their training ended** (almost four years for oil) and to stay in the competitive pack on every project.
- The **data pipeline is proven reproducible** from public sources in a fresh environment, byte-for-byte where the format allows and forecast-for-forecast where it doesn't.
- There is now a **reusable, automated deploy → audit → report chain** (tools/deploy/run_all.sh) that refuses to train, refuses to over-claim and fails on any integrity breach.
- **Five data-quality issues** that the earlier audit did not record are now documented, with their measured impact.
- **Sealed forecasts** exist for targets not yet published, so the first truly out-of-sample grades cannot be tuned after the fact.

9. Next steps

- **2026-11-05**: EIA publishes the week ending 2026-10-30. Run tools/deploy/score_exante.py (refuses if a sealed file changed) and tools/forward/forward_eval.py.
- Keep the forward test running: about 20 gas weeks or 60 power/crypto days are needed before any confidence interval is informative.
- Fix the forward-test label reference (finding 2) before November crypto days are scored.
- Review and merge PR #2.